BATCH & BUILD · STEM LEARNING SHEET · AGES 8–12

Battery Cars: Stored Energy on the Move

THE SCIENCE

A battery stores chemical energy. When you connect it into a circuit, which is a complete loop of wire, the chemicals react and push electricity through the wires.

The electricity reaches a motor. Inside, it flows through coils of wire sitting next to magnets. The magnets push and pull on the coils, making the motor shaft spin. That's electrical energy turning into kinetic energy, the energy of motion.

The spinning shaft turns the axle, either directly or through gears, and the wheels roll. Friction between the wheels and the ground lets the car grip and move forward. Friction in the axles slows it down. Good car designs keep the grip and cut down the rubbing.

A switch is a gate in the circuit. Open it and the loop breaks, so the car stops. Close it and the car goes.

KEY WORDS

Battery: stores energy as chemicals.

Circuit: a complete loop electricity can travel around.

Motor: turns electricity into spinning motion.

Axle: the rod the wheels turn on.

Friction: a force that slows things that rub together.

Switch: opens or closes a circuit.

Energy transfer: energy changing from one form to another.

EXPERIMENT – SURFACE RACE

Question: Which surface lets your car go fastest? Mark a 2-meter track with tape. Time your car on carpet, on a wood or tile floor, and on a smooth table, 3 runs each, with fresh batteries for every surface. Find the average time for each one.

Think about it: Which surface had the most friction, and how could you tell?

SAFETY

Use only the batteries that come with the kit. Never connect a battery's two ends directly with a wire, because it can get hot. Keep button batteries away from little kids, since swallowing one is dangerous. Ask an adult before changing the wiring.

CHECK YOUR UNDERSTANDING

1. What kind of energy does a battery store?
2. What does a motor do?
3. What happens when the circuit is broken?
4. Name one helpful and one unhelpful kind of friction in a car.
5. What does a switch do?

Name: _____

Battery Cars: Stored Energy on the Move

ANSWER KEY (FOR GROWN-UPS)

1. Chemical energy.
2. It turns electricity into spinning motion.
3. Electricity stops flowing, so the car stops.
4. Wheel grip helps; axle rubbing slows it down.
5. It opens or closes the circuit.

Name: _____